PCB Inspection Report

Timestamp	2026-04-04 16:03:42
Status	DEFECTIVE
Risk Level	HIGH
Failure Probability	70.06%

Detected Defects

#	Label	Confidence	BBox
1	Missing_hole	0.7954	[1192, 280, 1214, 301]
2	Missing_hole	0.6899	[802, 55, 848, 100]
3	Missing_hole	0.6693	[705, 55, 745, 98]
4	Missing_hole	0.649	[532, 274, 553, 297]
5	Missing_hole	0.5945	[108, 198, 131, 222]
6	Missing_hole	0.3384	[598, 56, 639, 98]
7	Missing_hole	0.3038	[954, 704, 973, 722]

Explanation

Technical Analysis: PCB Inspection Defect Report
1) Root Cause Analysis
The primary root cause for this "DEFECTIVE" status, driven by seven distinct "Missing_hole" defects, points directly to a critical failure within the PCB drilling process. The high confidence values (0.7954, 0.6899, 0.6693, 0.649) for the majority of the defects strongly indicate a physical absence of the intended drilled feature rather than a misidentification. Potential contributors include:
* **Drill Bit Failure/Wear:** Repeated missing holes could stem from a broken or severely worn drill bit, especially if these are small-diameter holes. The scatter of defects across the board (as suggested by varying bbox coordinates) could imply multiple drill hit failures or a systematic wear issue not triggering immediate machine stoppage.
* **Drill File (NC Data) Corruption:** Errors during the CAM data generation or transfer could lead to specific drill hits being omitted from the machine's instructions.
* **Drilling Machine Malfunction:** Issues with spindle runout, incorrect Z-axis calibration, or insufficient feed/speed settings for the material stack could prevent complete hole formation.
* **Panel Misregistration/Clamping Issues:** Inconsistent clamping or panel movement during drilling could cause entire regions to be skipped or misdrilled, though the scattered nature of these specific defects makes this less probable as a singular cause for all.
The lower confidence detections (0.3384, 0.3038) could represent partially drilled holes, holes obscured by drilling debris, or less distinct features at the detection threshold, warranting further microscopic validation to confirm their exact state but still indicating a drilling anomaly.
2) Operational Risks
The presence of seven "Missing_hole" defects elevates the operational risk to "HIGH" with a stated "Failure probability: 70.06%." This is justified because:
* **Catastrophic Electrical Failure:** If these missing holes are intended vias for signal or power routing, they will result in open circuits, rendering the PCB fundamentally non-functional at the

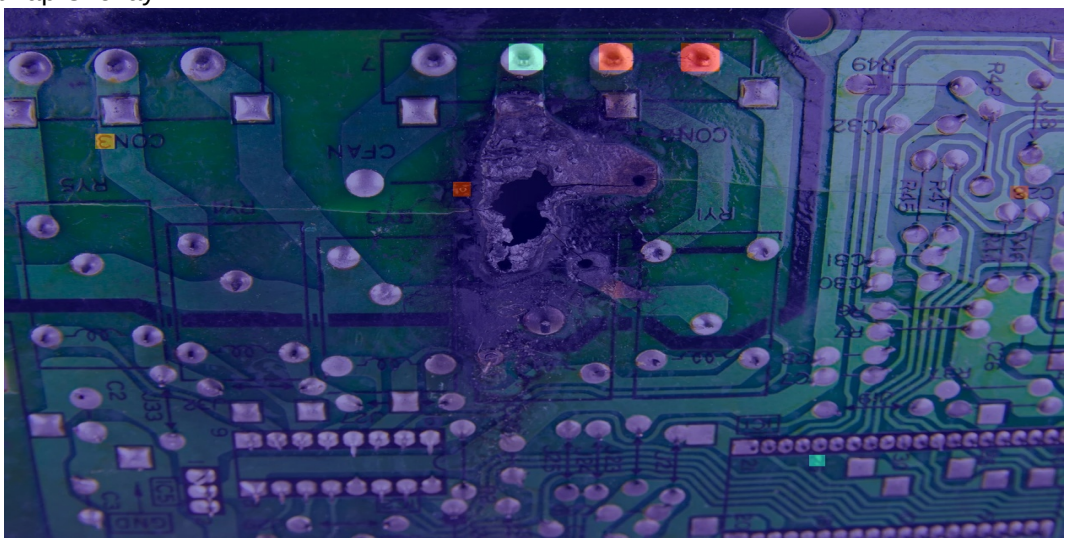
electrical test stage (e.g., continuity testing, In-Circuit Test (ICT)).
* **Assembly Impairment:** If these are component mounting holes, components cannot be properly placed, secured, or soldered, leading to immediate assembly line stoppages and significant yield loss. Forcing components onto partially drilled or non-existent holes can result in component damage or stress-induced latent failures.
* **Thermal Management Compromise:** If thermal vias are missing, critical heat dissipation paths will be absent, leading to localized overheating and premature component failure during operation.
* **Mechanical Integrity Loss:** For critical mounting or tooling holes, the structural integrity of the assembled product will be compromised.
3) Failure Prediction
Based on the "Missing_hole" defect type and "HIGH" risk level:
* **Immediate Functional Failure:** This PCB is highly likely to fail initial electrical tests (e.g., bare board test, ICT) due to open circuits or short circuits if components are forced onto improperly drilled pads. It will certainly fail assembly if component leads cannot pass through the board.
* **Assembly Yield Impact:** Production of any higher-level assembly using this PCB will be immediately impacted. The board will be rejected, preventing further assembly steps.
* **No Latent Failure Mode for *Missing*:** For genuinely missing holes, a latent failure mode is unlikely as the defect presents as an immediate functional failure. However, if the lower confidence defects represent partially drilled holes, these could potentially lead to intermittent connections under mechanical stress or thermal cycling, degrading reliability over time.
4) Repair Explanation
Repairing a PCB with seven "Missing_hole" defects is generally **not recommended and not feasible for production units**, especially when dealing with critical vias or plated through-holes.
* **Scrap & Replace:** The most robust and reliable solution for a board exhibiting this number and type of critical defect is to scrap the unit and replace it with a defect-free board.
* **Manual Rework Risks:** Attempting to manually drill missing holes post-fabrication introduces significant risks:
* **Plating Integrity:** For vias or plated holes, drilling after fabrication will destroy any existing plating in the hole barrel, compromising electrical connectivity and solderability. Re-plating is not a viable option on a finished PCB.
* **Layer Damage:** Manual drilling can damage inner layers, create burrs, or introduce debris, leading to new electrical shorts or opens.
* **Dimensional Accuracy:** Achieving the precise diameter and location of a drilled hole manually is extremely challenging, impacting component fit and electrical clearance.
* **Reliability Compromise:** Any manual rework on such fundamental defects severely compromises the long-term reliability and manufacturability of the PCB.
5) Engineering Insights
The consistent "Missing_hole" defect type across multiple locations demands an immediate and thorough investigation into the PCB fabrication facility's drilling operations.
* **Process Control Monitoring:** Implement or review Statistical Process Control (SPC) for drilling. Focus on drill bit lifespan, machine spindle maintenance, and routine calibration.
* **CAD/CAM Data Verification:** Conduct a rigorous audit of the NC drill files and compare them against the original design intent (Gerber files, drill legend) to rule out data corruption or translation errors.
* **Drill Tool Management:** Enhance drill bit wear detection mechanisms and ensure strict adherence to drill bit replacement schedules based on material, hole count, and diameter. The presence of lower confidence defects suggests some holes may be partially drilled, potentially indicating insufficient drill bit penetration or debris accumulation.
* **AOI Calibration:** While the AOI effectively caught these defects, review the AOI calibration parameters, particularly for edge cases or smaller features, to ensure consistent and accurate detection of partially drilled holes or subtle anomalies in addition to outright missing holes.

* **Yield Impact Analysis:** Quantify the yield loss from this specific defect type to prioritize corrective actions and justify investment in process improvements. This is not an isolated incident if 7 such defects appear on a single board.

Repair Suggestions

- Missing_hole: Replace the missing component and verify orientation and BOM value before reflow.

Inspection Images

Original Image

Heatmap Overlay